

Qualification Pack



Textile Waste Recycling Machine Operator

Electives: Cleaning and Processing/ Shredding

QP Code: TSC/Q0121

Version: 1.0

NSQF Level: 2

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TSC/Q0121: Textile Waste Recycling Machine Operator

Brief Job Description

A Textile Waste Recycling Machine Operator is responsible for collection, sorting and shredding of textile waste under supervision by maintaining health, safety and security, and complying with the workplace requirements in the textile waste processing sector.

Personal Attributes

A Textile Waste Recycling Machine Operator should have physicals skills like dexterity, hand-eye leg coordination, motor skills, strength, stamina, ability to lift and carry heavy loads, and perform tasks requiring physical effort. A Textile Waste Recycling Machine Operator should also have sensory skills like hearing ability and keen eye focus.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TSC/N9015: Follow machine, safety, and organizational guidelines in textile sector](#)
2. [TSC/N9016: Follow teamwork, adaptability, and communication guidelines in textile sector](#)
3. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(*mandatory to select at least one*):

Elective 1: Cleaning and Processing

A Waste Recycling Machine Operator is responsible for the cleaning, cutting, and preparation of textile waste materials for recycling or repurposing. This role is crucial in ensuring that textile waste is processed efficiently and effectively, contributing to the sustainability efforts within the textile industry.

1. [TSC/N0135: Carryout textile waste cleaning and processing](#)

Elective 2: Shredding

A Waste Recycling Machine Operator - Shredding is responsible for operating and maintaining shredding machines that process textile waste into smaller, manageable pieces for recycling or disposal. This role is essential in ensuring that textile waste is efficiently shredded, contributing to the overall waste management and recycling processes.

1. [TSC/N0136: Carryout textile waste shredding operation](#)

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Qualification Pack (QP) Parameters

Sector	Textile
Sub-Sector	Spinning – Textiles
Occupation	Pre Spinning Operations, Preparatory - Weaving
Country	India
NSQF Level	2
Credits	14
Aligned to NCO/ISCO/ISIC Code	8151.9900
Minimum Educational Qualification & Experience	Ability to read and write
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	17/12/2027
NSQC Approval Date	17/12/2024
Version	1.0
Reference code on NQR	QG-02-TX-03439-2024-V1-TSC
NQR Version	1.0

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TSC/N9015: Follow machine, safety, and organizational guidelines in textile sector

Description

This unit provides performance criteria, knowledge and skills required to follow machine, safety and organizational guidelines in textile sector.

Scope

The scope covers the following :

- Maintaining the work area, tools and machines
- Greening and energy conservation in textile sector
- Health, safety and response to emergencies at textile sector
- Organizational standards and policies

Elements and Performance Criteria

Maintaining the work area, tools and machines

To be competent, the user/individual on the job must be able to:

- PC1.** handle materials, machinery, equipment and tools as per standard procedure
- PC2.** use appropriate material handling equipment and tools as per standard procedure
- PC3.** keep the equipment, machine and work area clean using appropriate cleaning tools as per standard procedure
- PC4.** undertake minor routine maintenance of equipment and tools as per standard maintenance procedure
- PC5.** maintain record for defective and unsafe equipment and tools
- PC6.** verify that machine guards are in place as per standard specifications
- PC7.** follow specified ergonomics for the assigned job role in textile sector
- PC8.** collect and store worn-out spare parts at specified location
- PC9.** report the condition of worn out parts as per standard procedure

Greening and energy conservation in textile sector

To be competent, the user/individual on the job must be able to:

- PC10.** segregate wastes such as recyclable, non-recyclable, hazardous as per standard protocol
- PC11.** optimize usage of material and resources including water, electricity in various tasks
- PC12.** switch off the machines and lights when not in use

Health, safety and response to emergencies at textile sector

To be competent, the user/individual on the job must be able to:

- PC13.** use Personal Protective Equipment (PPEs) like body protector, ear plugs, nose mask, head cap, etc. as per guidelines
- PC14.** identify abnormal sounds emanating from faulty or worn out machine parts and take appropriate action
- PC15.** avoid dependency on any type of intoxicants

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- PC16.** maintain social distance as per the instruction at workplace
- PC17.** report hazardous material to superiors at workplace
- PC18.** use the various appropriate fire extinguishers on different types of fires correctly
- PC19.** follow the specified steps in case of electricity failure
- PC20.** lift heavy objects using correct lifting procedures
- PC21.** recall emergency exits, safe spots, etc. of workplace
- PC22.** practice mock drills and evacuation procedures organized by industry
- PC23.** assist others to reach to safe spots in emergency situations
- PC24.** provide basic first aid for injury to peers and report to superiors
- PC25.** interpret different signs, alarms and take action appropriately
- PC26.** follow the guidelines while working in hazards atmosphere
- PC27.** assist in designing the safety plans with peers and superiors
- PC28.** follow the approved safety plans at workplace

Organizational standards and policies

To be competent, the user/individual on the job must be able to:

- PC29.** perform assigned duties as per organization's protocol within scheduled time period
- PC30.** follow organization policies, quality standards, rules and regulations for working in textile sector
- PC31.** motivate colleagues to follow operational guidelines of organization
- PC32.** wear specified uniform and follow etiquette as per standard guidelines for the textile sector
- PC33.** maintain hygienic working atmosphere as per protocol of the textile sector
- PC34.** submit lost and found articles as per standard protocol

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational standard procedures, quality standards, rules, codes, policies and safety standards in the textile sector
- KU2.** different type of tools and equipment used in textile sub- sector and their specifications and operating procedures
- KU3.** safe handling procedure of tools and equipment
- KU4.** the importance of displays and written instructions for the allocated machines
- KU5.** dos and donts specific to the assigned work responsibilities
- KU6.** protocol for minimizing the wastage of material, effort and time
- KU7.** organization's formats and procedures for reporting production, defects, faults, material/tool requisition and quality parameters and task completed for assigned job
- KU8.** schedule for cleaning and waste collection for the assigned job role
- KU9.** importance and standard procedure for disposal of soft, hard, non-hazardous and hazardous wastes and materials
- KU10.** available types of material handling equipment and handling methods used in the textile sector

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- KU11.** hazards of unsafe workplace conditions and procedures in the textile industry and methods to avoid hazards
- KU12.** various types of fire extinguishers
- KU13.** importance of stable mental condition in case of emergency
- KU14.** correct work posture and importance of ergonomics for the assigned job role
- KU15.** organizational quality systems like quality circle, 5S, ISO, SA, etc. followed in the textile sector
- KU16.** importance of following work wear standards, behavioral protocols and etiquette in the textile sector
- KU17.** importance of energy conservation through proper maintenance schedule in the textile sector
- KU18.** procedures and formats for reporting lost and found material
- KU19.** different types of alarms and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and comprehend policies in notice boards displayed in the workplace
- GS2.** minimize the resource consumption for the assigned task
- GS3.** solve basic arithmetic calculations related to assigned job role
- GS4.** recognize and differentiate colors of materials used in textile sector
- GS5.** lift specified materials for the allotted task using prescribed ergonomic position
- GS6.** exhibit motor skill required for the allotted task

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintaining the work area, tools and machines</i>	10	10	-	6
PC1. handle materials, machinery, equipment and tools as per standard procedure	-	-	-	-
PC2. use appropriate material handling equipment and tools as per standard procedure	-	-	-	-
PC3. keep the equipment, machine and work area clean using appropriate cleaning tools as per standard procedure	-	-	-	-
PC4. undertake minor routine maintenance of equipment and tools as per standard maintenance procedure	-	-	-	-
PC5. maintain record for defective and unsafe equipment and tools	-	-	-	-
PC6. verify that machine guards are in place as per standard specifications	-	-	-	-
PC7. follow specified ergonomics for the assigned job role in textile sector	-	-	-	-
PC8. collect and store worn-out spare parts at specified location	-	-	-	-
PC9. report the condition of worn out parts as per standard procedure	-	-	-	-
<i>Greening and energy conservation in textile sector</i>	7	10	-	6
PC10. segregate wastes such as recyclable, non-recyclable, hazardous as per standard protocol	-	-	-	-
PC11. optimize usage of material and resources including water, electricity in various tasks	-	-	-	-
PC12. switch off the machines and lights when not in use	-	-	-	-
<i>Health, safety and response to emergencies at textile sector</i>	32	40	-	28

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. use Personal Protective Equipment (PPEs) like body protector, ear plugs, nose mask, head cap, etc. as per guidelines	-	-	-	-
PC14. identify abnormal sounds emanating from faulty or worn out machine parts and take appropriate action	-	-	-	-
PC15. avoid dependency on any type of intoxicants	-	-	-	-
PC16. maintain social distance as per the instruction at workplace	-	-	-	-
PC17. report hazardous material to superiors at workplace	-	-	-	-
PC18. use the various appropriate fire extinguishers on different types of fires correctly	-	-	-	-
PC19. follow the specified steps in case of electricity failure	-	-	-	-
PC20. lift heavy objects using correct lifting procedures	-	-	-	-
PC21. recall emergency exits, safe spots, etc. of workplace	-	-	-	-
PC22. practice mock drills and evacuation procedures organized by industry	-	-	-	-
PC23. assist others to reach to safe spots in emergency situations	-	-	-	-
PC24. provide basic first aid for injury to peers and report to superiors	-	-	-	-
PC25. interpret different signs, alarms and take action appropriately	-	-	-	-
PC26. follow the guidelines while working in hazards atmosphere	-	-	-	-
PC27. assist in designing the safety plans with peers and superiors	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. follow the approved safety plans at workplace	-	-	-	-
<i>Organizational standards and policies</i>	16	20	-	10
PC29. perform assigned duties as per organization's protocol within scheduled time period	-	-	-	-
PC30. follow organization policies, quality standards, rules and regulations for working in textile sector	-	-	-	-
PC31. motivate colleagues to follow operational guidelines of organization	-	-	-	-
PC32. wear specified uniform and follow etiquette as per standard guidelines for the textile sector	-	-	-	-
PC33. maintain hygienic working atmosphere as per protocol of the textile sector	-	-	-	-
PC34. submit lost and found articles as per standard protocol	-	-	-	-
NOS Total	65	80	-	50

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National Occupational Standards (NOS) Parameters

NOS Code	TSC/N9015
NOS Name	Follow machine, safety, and organizational guidelines in textile sector
Sector	Textile
Sub-Sector	Generic - Textiles Handloom
Occupation	Generic - Textiles & Handloom
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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TSC/N9016: Follow teamwork, adaptability, and communication guidelines in textile sector

Description

This unit provides performance criteria, knowledge and skills required to follow team work, communication and adaptability in textile sector

Scope

The scope covers the following :

- Teamwork and communication
- Adaptability

Elements and Performance Criteria

Teamwork, trust and communication

To be competent, the user/individual on the job must be able to:

- PC1.** contribute to create a positive work environment in the team
- PC2.** carry out tasks as per instructions received from superiors
- PC3.** contribute to team work as per allocated responsibility to complete the task by using appropriate tools and methods
- PC4.** build trust with team mates and superiors
- PC5.** implement the ideas after superior's approval at work place
- PC6.** communicate clearly with the team members as per standard protocol
- PC7.** use suggested hand signs, vocal sound signals to convey the information in the production area
- PC8.** listen effectively to the ideas and concerns of the peers
- PC9.** use correct and respectful terms while communicating as per industry policy
- PC10.** express views proactively and effectively
- PC11.** make efforts to resolve difference of opinion with superiors and team members
- PC12.** report to superior for problems identified in assigned duty
- PC13.** report the daily performance to superior in prescribed manner and formats

Adaptability

To be competent, the user/individual on the job must be able to:

- PC14.** adapt to flexible work environment for the assigned task
- PC15.** adapt to work with various members of different ethnicity, gender and PwD without biases
- PC16.** consider opinions of colleagues, fitters, superiors for the assigned task
- PC17.** plan the work-routine within the limits of the responsibility
- PC18.** adopt new ideas after due approval from superior for improving the productivity

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** importance of teamwork and discipline
- KU2.** limits and responsibilities for the assigned duties in the textile sector
- KU3.** possible conflicts in the assigned job role and methods to resolve the same
- KU4.** importance of teamwork, group discussions and healthy work environment
- KU5.** importance of reporting as per the standard protocol
- KU6.** hierarchy of communication and communication etiquettes in the textile sector
- KU7.** protocol for communication with different ethnicity, gender and PwD
- KU8.** report and grievance submission formats
- KU9.** importance of hand, vocal sound signals in the textile sector

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and comprehend written instructions
- GS2.** listen effectivity for the instructions
- GS3.** coordinate with team mates for the allotted tasks
- GS4.** use and comprehend prescribed voice and hand signals in the textile production area
- GS5.** fill forms and prepare reports such as production report, material requisition forms, leave application, etc. as per standard formats

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Teamwork, trust and communication</i>	10	15	-	5
PC1. contribute to create a positive work environment in the team	-	-	-	-
PC2. carry out tasks as per instructions received from superiors	-	-	-	-
PC3. contribute to team work as per allocated responsibility to complete the task by using appropriate tools and methods	-	-	-	-
PC4. build trust with team mates and superiors	-	-	-	-
PC5. implement the ideas after superior's approval at work place	-	-	-	-
PC6. communicate clearly with the team members as per standard protocol	-	-	-	-
PC7. use suggested hand signs, vocal sound signals to convey the information in the production area	-	-	-	-
PC8. listen effectively to the ideas and concerns of the peers	-	-	-	-
PC9. use correct and respectful terms while communicating as per industry policy	-	-	-	-
PC10. express views proactively and effectively	-	-	-	-
PC11. make efforts to resolve difference of opinion with superiors and team members	-	-	-	-
PC12. report to superior for problems identified in assigned duty	-	-	-	-
PC13. report the daily performance to superior in prescribed manner and formats	-	-	-	-
<i>Adaptability</i>	5	5	-	5
PC14. adapt to flexible work environment for the assigned task	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. adapt to work with various members of different ethnicity, gender and PwD without biases	-	-	-	-
PC16. consider opinions of colleagues, fitters, superiors for the assigned task	-	-	-	-
PC17. plan the work-routine within the limits of the responsibility	-	-	-	-
PC18. adopt new ideas after due approval from superior for improving the productivity	-	-	-	-
NOS Total	15	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TSC/N9016
NOS Name	Follow teamwork, adaptability, and communication guidelines in textile sector
Sector	Textile
Sub-Sector	Generic - Textiles Handloom
Occupation	Generic - Textiles & Handloom
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026

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TSC/N0135: Carryout textile waste cleaning and processing

Description

This NOS unit about performance criteria, knowledge and understanding and generic skills required for executing necessary techniques to clean, cut, and prepare the textile waste materials for recycling or repurposing.

Scope

The scope covers the following :

- Cleaning and Cutting the Textile Waste
- Segregation and Quality Inspection

Elements and Performance Criteria

Cleaning and Cutting the Textile Waste

To be competent, the user/individual on the job must be able to:

- PC1.** Collect and open the received waste bundles/ bales for cleaning/ processing activity
- PC2.** Remove dirt, debris, and contaminants from textile waste using appropriate cleaning methods and equipment
- PC3.** Use cutting tools or machinery to trim and cut textile waste into specified sizes and shapes
- PC4.** Follow guidelines and specifications for cutting to maintain uniformity and maximize material usage
- PC5.** Store the removed trims/ accessories such as buttons, zippers, etc. at the designated area
- PC6.** Clean the textile waste using prescribed method further processing or recycling

Segregation and Quality Inspection

To be competent, the user/individual on the job must be able to:

- PC7.** Sort textiles by material type, color, and quality before and after cleaning and cutting
- PC8.** Identify and separate reusable, recyclable, and non-recyclable materials.
- PC9.** Inspect textile waste after cleaning and cutting to its quality standards for further processing
- PC10.** Report any issues or defects to supervisors for necessary action
- PC11.** Store the inspected waste at designated area for further processing
- PC12.** Maintain the records of cleaned and cut textile waste, including quantities and types

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of textile waste such as pre-consumer, post-consumer
- KU2.** sources and composition of textile wastes
- KU3.** environmental impact of textile waste
- KU4.** textile waste collection and sorting methods

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- KU5.** various recycling process such as mechanical (shredding, carding, spinning) and chemical recycling process (solvent-based, thermal)
- KU6.** tools available for waste classification such as Near-Infrared (NIR) spectroscopy, and optical, manual methods for sorting and segregation
- KU7.** importance of recycling and waste reduction
- KU8.** awareness of the environmental consequences of improper waste disposal
- KU9.** various types of textiles such as cotton, polyester, wool, silk, and synthetic blends
- KU10.** processes involved in recycling different textiles
- KU11.** safety procedures to prevent accidents and injuries while handling textile waste
- KU12.** appropriate PPE for cleaning and cutting of textile waste materials
- KU13.** compliance requirements for handling and disposing of textile waste
- KU14.** various records for capturing the collected textile waste

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** perform basic arithmetic calculations for records keeping
- GS2.** communicate effectively with team members and supervisors
- GS3.** work cooperatively with colleagues and supervisors to ensure smooth operations
- GS4.** read and understand work orders, safety instructions, and regulatory documents
- GS5.** adapt to various procedures suggested by management as required

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cleaning and Cutting the Textile Waste</i>	75	150	-	25
PC1. Collect and open the received waste bundles/ bales for cleaning/ processing activity	-	-	-	-
PC2. Remove dirt, debris, and contaminants from textile waste using appropriate cleaning methods and equipment	-	-	-	-
PC3. Use cutting tools or machinery to trim and cut textile waste into specified sizes and shapes	-	-	-	-
PC4. Follow guidelines and specifications for cutting to maintain uniformity and maximize material usage	-	-	-	-
PC5. Store the removed trims/ accessories such as buttons, zippers, etc. at the designated area	-	-	-	-
PC6. Clean the textile waste using prescribed method further processing or recycling	-	-	-	-
<i>Segregation and Quality Inspection</i>	50	130	-	20
PC7. Sort textiles by material type, color, and quality before and after cleaning and cutting	-	-	-	-
PC8. Identify and separate reusable, recyclable, and non-recyclable materials.	-	-	-	-
PC9. Inspect textile waste after cleaning and cutting to its quality standards for further processing	-	-	-	-
PC10. Report any issues or defects to supervisors for necessary action	-	-	-	-
PC11. Store the inspected waste at designated area for further processing	-	-	-	-
PC12. Maintain the records of cleaned and cut textile waste, including quantities and types	-	-	-	-
NOS Total	125	280	-	45

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TSC/N0135
NOS Name	Carryout textile waste cleaning and processing
Sector	Textile
Sub-Sector	
Occupation	Pre Spinning Operations, Preparatory - Weaving
NSQF Level	2
Credits	5
Version	1.0
Last Reviewed Date	17/12/2024
Next Review Date	17/12/2027
NSQC Clearance Date	17/12/2024

Qualification Pack

TSC/N0136: Carryout textile waste shredding operation

Description

This NOS unit about performance criteria, knowledge and understanding and generic skills required for executing necessary techniques to operate and maintain shredding machines that process textile waste into smaller, manageable pieces for recycling activity.

Scope

The scope covers the following :

- Operate the shredding machine
- Maintenance and Quality Control in shredding

Elements and Performance Criteria

Operate the shredding machine

To be competent, the user/individual on the job must be able to:

- PC1.** Load the cleaned textile waste materials onto the shredding machine
- PC2.** Set up, operate, and monitor shredding machines to process textile waste
- PC3.** Adjust machine settings to ensure optimal performance and efficiency
- PC4.** Feed textile waste into the shredding machine, ensuring a continuous and safe workflow

Maintenance and Quality Control in shredding

To be competent, the user/individual on the job must be able to:

- PC5.** Troubleshoot and resolve minor mechanical issues in the allotted shredding machine within the limits of responsibility
- PC6.** Inspect shredded materials according to specified size and quality standards
- PC7.** Identify and remove contaminants or non-shreddable items during the shredding operation
- PC8.** Unload the shredded materials, by following standard handling and storage methods

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of textile waste such as pre-consumer, post-consumer
- KU2.** sources and composition of textile wastes
- KU3.** environmental impact of textile waste
- KU4.** textile waste collection and sorting methods
- KU5.**
 - various recycling process such as mechanical (shredding, carding, spinning) and chemical
 - recycling process (solvent-based, thermal)
- KU6.**
 - tools available for waste classification such as Near-Infrared (NIR) spectroscopy, and optical,
 - manual methods for sorting and segregation
- KU7.** importance of recycling and waste reduction

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- KU8.** awareness of the environmental consequences of improper waste disposal
- KU9.** various types of textiles such as cotton, polyester, wool, silk, and synthetic blends
- KU10.** contaminated or non-recyclable textiles
- KU11.** processes involved in recycling different textiles
- KU12.** safety procedures to prevent accidents and injuries while handling textile waste
- KU13.** appropriate PPE for different tasks and materials
- KU14.** compliance requirements for handling and disposing of textile waste
- KU15.** various records for capturing the collected textile waste
- KU16.** shredding machine operating principle and maintenance schedules blade sharpening, lubrications, etc.
- KU17.** various shredding mechanisms hammer mills, granulators, etc
- KU18.** various machine settings involved in shredding operation blade configuration, speed, and feed rate, etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write instructions for communication, production data entry, etc.
- GS2.** perform basic arithmetic calculations for records keeping
- GS3.** communicate effectively with team members and supervisors
- GS4.** work cooperatively with colleagues and supervisors to ensure smooth operations
- GS5.** read and understand work orders, safety instructions, and regulatory documents
- GS6.** adapt to various procedures suggested by management as required

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the shredding machine</i>	75	150	-	25
PC1. Load the cleaned textile waste materials onto the shredding machine	-	-	-	-
PC2. Set up, operate, and monitor shredding machines to process textile waste	-	-	-	-
PC3. Adjust machine settings to ensure optimal performance and efficiency	-	-	-	-
PC4. Feed textile waste into the shredding machine, ensuring a continuous and safe workflow	-	-	-	-
<i>Maintenance and Quality Control in shredding</i>	50	130	-	20
PC5. Troubleshoot and resolve minor mechanical issues in the allotted shredding machine within the limits of responsibility	-	-	-	-
PC6. Inspect shredded materials according to specified size and quality standards	-	-	-	-
PC7. Identify and remove contaminants or non-shreddable items during the shredding operation	-	-	-	-
PC8. Unload the shredded materials, by following standard handling and storage methods	-	-	-	-
NOS Total	125	280	-	45

Qualification Pack

National Occupational Standards (NOS) Parameters

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Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

The assessment for the theory part will be based on knowledge bank of questions created by the SSC. Individual assessment agencies will create unique question papers for theory part for each candidate at

each examination/training center (as per assessment criteria below). Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 50

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(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N9015.Follow machine, safety, and organizational guidelines in textile sector	65	80	-	50	195	25
TSC/N9016.Follow teamwork, adaptability, and communication guidelines in textile sector	15	20	-	10	45	6
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	9
Total	100	130	-	60	290	40

Elective: 1 Cleaning and Processing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N0135.Carryout textile waste cleaning and processing	125	280	-	45	450	60
Total	125	280	-	45	450	60

Elective: 2 Shredding

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TSC/N0136.Carryout textile waste shredding operation	125	280	-	45	450	60

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	125	280	-	45	450	60

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.